

Computer Networking Short Notes

1. OSI Model

7 Layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.

2. TCP/IP Model

4 Layers: Network Interface, Internet, Transport, Application.

3. TCP vs UDP

TCP: Reliable, connection-oriented.

UDP: Fast, connectionless, no guarantee.

4. IP Addressing

IPv4: 32-bit address. IPv6: 128-bit address.

Used to uniquely identify devices on a network.

5. DNS

Domain Name System converts domain names to IP addresses.

6. Routing & Switching

Router: Connects different networks.

Switch: Connects devices in same network.

7. Protocols

HTTP/HTTPS: Web communication.

FTP: File transfer.

SMTP: Email sending.

8. Network Security

Firewall, Encryption, VPN used for security.

